

DENSITY ORDINATES OF POPULATION PERCENTILES AND ORDER STATISTICS

Gary J. Manas¹

Department of Statistics and Actuarial Science, University of the Witwatersrand, P.O. Box 2050, Johannesburg, 2050

Key words: Entropy; likelihood; frequency moment estimator; sufficient statistic.

Summary: This paper introduces and suggests applications for the function relating percentile ranks to density ordinates in continuous populations. In particular, it provides the (almost sure) limiting value of the density ordinate of the i -th order statistic of a random sample from the population. The function also gives a likelihood based estimation procedure which asymptotically gives the frequency moment estimators. Examples are taken from the exponential, double exponential, Cauchy, logistic and Pareto distributions.

1. Introduction

Let X denote a continuous random variable with distribution function $F_X(x)$ and density $f_X(x) = F'_X(x)$. In this paper, some properties and inferential applications are established for the function $\tau_X(u) = f_X(F_X^{-1}(u))$, where $0 < u < 1$. In particular, we deduce asymptotic properties of the sequence of random variables, $\{f_X(Y_{i,n})\}_{i=1}^n$, where $Y_{1,n} < Y_{2,n} < \dots < Y_{n,n}$ denote the order statistics associated with a random sample of size n , $X_1, \dots, X_n$ from X .

¹ Gary Manas died on 13 August 2001.

Consider, initially, the i -th density ordinate associated with X as the random variable, defined by

$$\xi_{i,n} = f_X(Y_{i,n}), \quad (1.1)$$

for $i = 1, \dots, n$. It is important to note that the distribution of $\xi_{i,n}$ is uniquely determined by X for given n . Conversely, it follows by the uniqueness of f_X that if an X exists for a given $\xi_{i,n}$, then it is unique up to location shifts.

Despite some interesting properties and applications, it appears that there is no literature on the sequence, $\{\xi_{i,n}\}_{i=1}^n$ except for an implicit reference to asymptotic expansions of moments of order statistics in terms of their corresponding population percentiles (David, 1981 p. 80). Properties of the ordinates of the distribution function, i.e.

$$U_{i,n} = F_X(Y_{i,n}), \quad (1.2)$$

however, are well established, and (1.2) defines the so-called 'probability integral' transformation. (See e.g. David, 1981, p.80; Harter, 1988, p.2101.) In particular, it is well known that $U_{i,n}$ has the Beta $(i, n - i + 1)$ distribution, from which it follows that

$$E(U_{i,n}) = \frac{i}{n+1} \quad (1.3)$$

and

$$\text{Var}(U_{i,n}) = \frac{i(n-i+1)}{(n+2)(n+1)^2}. \quad (1.4)$$

The probability integral transformation has proved most useful in inference (see e.g. Barnett, 1975; Harter, 1984 and Harter and Weigand, 1985).

In the present work, a result corresponding to (1.3) is established for the density ordinates, τ_X . The paper proceeds with the main results and some corollaries and applications, particularly in likelihood estimation. A section

of examples follows in which the results are applied to several well-known distributions, including the exponential, Pareto, Cauchy and logistic. The purpose of these examples is to illustrate properties of the so-called 'density ordinate' function upon which the theory is based.

2. Preliminary Remarks and Results

Since F_X is strictly increasing over the interval for which $f_X > 0$, it follows that F_X^{-1} exists on this interval with

$$F_X^{-1}(F_X(x)) = x \quad (2.1)$$

for $-\infty < x < \infty$ and

$$F_X(F_X^{-1}(u)) = u \quad (2.2)$$

for $0 < u < 1$. Now set $u = i/(n+1)$ and let $n, i \rightarrow \infty$ in such a way that u remains constant. Clearly $F_X^{-1}(u)$ is the $100i/(n+1)$ -th percentile of X and it is well known that $Y_{i,n}$ approaches this percentile, almost surely, as $n \rightarrow \infty$. It thus follows that the limiting situation in (2.2) gives

$$U_{i,n} \rightsquigarrow \frac{i}{n+1}. \quad (2.3)$$

(The notation ' $\rightsquigarrow$ ' denotes that the ratio and difference of the two sides tend to 1 and 0, respectively, almost surely, under this limiting condition on i and n . It is used for convenience to permit putting n on both sides of the asymptotic results.) Note that (2.3) is a well-known result which can also be deduced from (1.3) and the strong law of large numbers.

Next, a result corresponding to (2.3) for the density function, f_X is similarly obtained by considering the limit with respect to the first derivative of (2.2). From the chain rule,

$$F_X^{-1'}(u)f_X(F_X^{-1}(u)) = 1, \quad (2.4)$$

for $0 < u < 1$. This gives the

THEOREM:

$$\xi_{i,n} \rightsquigarrow \frac{1}{F_X^{-1'}(i/(n+1))} = \tau_X\left(\frac{i}{n+1}\right), \quad (2.5)$$

where $F_X^{-1'}$ denotes the first derivative of F_X^{-1} .

PROOF:

This follows since

$$\tau_X(u) = f_X(F_X^{-1}(u)) = \frac{1}{F_X^{-1'}(u)}, \quad (2.6)$$

for $0 < u < 1$. The alternate forms of τ_X in (2.6) are obtained by respective differentiation of (2.1) and (2.2).

From (2.6) it follows that

$$F_X^{-1}(u) = \int \frac{du}{\tau_X(u)} + c, \quad (2.7)$$

where c is a constant of integration, the value of which determines a pure location parameter of X . Note that (2.7) defines an indefinite integral which determines $F_X^{-1}(u)$, and hence F_X . In addition (2.7) implies that, excluding location, τ_X characterises the distribution of X . It follows that for given τ_X population models can be constructed from (2.7).

3. Properties of τ_X

3.1

The Jacobian, J , of the probability integral transformation, $u = F_X(x)$ equals $dx/du = 1/\tau_X(u)$.

This follows from (2.4) and the definition of J .

3.2

If U is uniformly distributed on $(0,1)$ then $\tau_X(U)$ and $f_X(X)$ have the same distribution.

This follows since $F^{-1}(U)$ is distributed as X .

3.3

The probability integral transformation provides the following alternative integral representation for population moments (about 0) of X :

$$\mu'_r = E(X^r) = \int_0^1 \{F_X^{-1}(u)\}^r du,$$

for $r = 1, 2, \dots$, which is useful on account of the bounded interval for the integral operator.

3.4

The frequency moments of order s are given by

$$E(f_X^s(X)) = \int_{-\infty}^{\infty} f_X^{s+1}(x) dx = \int_0^1 \tau_X^s(u) du.$$

The frequency moments have proved useful in inference (Manas, 1997).

3.5

The entropy of X is

$$H_X = -E(\log f_X(X)) = -\int_0^1 \log \tau_X(u) du.$$

3.6

If $X \in (a, b)$ defines the range of X , then $\tau_X(0) = a$ and $\tau_X(1) = b$. It follows that if the density of X has a right tail, then $\tau_X(1) = 0$. A left tail similarly implies that $\tau_X(0) = 0$.

3.7

The hazard rate or survival function of a non-negative X is given by

$$\lambda(x) = \frac{f_X(x)}{1 - F_X(x)} = \frac{\tau_X(u)}{1 - u}$$

From this it follows that if X is memoryless (exponential) then λ is constant with $\tau_X(u) \propto (1 - u)$. It also provides insights into parameterizing memoryful waiting times. For example, if $\tau_X(u) = (1 - u)^\alpha$, where α denotes a constant, positive parameter, then X is memoryless for $\alpha = 1$, long tailed or retentive for $\alpha > 1$ and short tailed and bounded (memory deficient) for $0 < \alpha < 1$. This τ_X produces the well-known generalized Pareto distribution. (See Figure 1 for graphs of F_X^{-1} , τ_X and f_X for $\alpha = 0.5, 1$ and 2 . Note how the tail is affected for these values of α .)

4. Asymptotic density ordinates

THEOREM:

A real number, x , satisfies $F_X(x) = i/(n+1)$ for some n and $i = 1, \dots, n$ if and only if

$$f_X(x) = \tau_X\left(\frac{i}{n+1}\right). \quad (4.1)$$

(This theorem is proved in the appendix.)

It is clear from (2.6) that the sequence $\{\tau_X(i/(n+1))\}_{i=1}^n$ coincides with f_X over an increasing sequence of abscissae, $x_1 < x_2 < \dots < x_n$, where x_i is the $100i/(n+1)$ -th percentile of X . Since $f_X(x_i) = \tau_X(i/(n+1))$ on this sequence of percentiles, τ_X effectively provides an alternate representation of the density, f_X . An immediate consequence of (4.1) is that $P(x_i < X <$

$x_{i+1}) = 1/(n+1)$ for $i = 1, \dots, n-1$ so that asymptotically, density ordinates, like order statistics, divide the area under f_X into equal portions.

Also, from theorem (2.5) and the strong law of large numbers, it follows that

$$E(\xi_{i,n}) \leadsto \tau_X\left(\frac{i}{n+1}\right), \quad (4.2)$$

where the limit refers to the above situation for i, n , but ordinary convergence of series (and not almost sure convergence) can apply.

COROLLARY:

$$\frac{n!}{(i-1)!(n-i)!} \int_0^1 \tau_X(u) u^{i-1} (1-u)^{n-i} du \leadsto \tau_X\left(\frac{i}{n+1}\right), \quad (4.3)$$

since, by definition,

$$E(\xi_{i,n}) = \int_{-\infty}^{\infty} f_X(x) f_{Y_{i,n}}(x) dx, \quad (4.4)$$

where the density function of $Y_{i,n}$ is

$$f_{Y_{i,n}}(x) = \frac{n!}{(i-1)!(n-i)!} f_X(x) \{F_X(x)\}^{i-1} \{1-F_X(x)\}^{n-i}. \quad (4.5)$$

(4.3) follows by substituting $u = F_X(x)$ into (4.4) and applying (4.5).

5. Empirical Density Ordinates

From the definition of the derivative it follows that

$$f_X(x) = \lim_{y \rightarrow x, 0+} \frac{F_X(y) - F_X(x)}{y - x}, \quad (5.1)$$

where the limit is taken through positive values of $y - x$. Now put $x = Y_{i,n}$, $y = Y_{i+1,n}$ and let $n \rightarrow \infty$. Using (2.3), (5.1) gives

$$\xi_{i,n} \rightsquigarrow \frac{1}{(n+1)\{Y_{i+1,n} - Y_{i,n}\}} \quad (5.2)$$

for $i = 1, \dots, n-1$. Thus from (2.5),

$$\hat{\tau}_X\left(\frac{i}{n+1}\right) \approx \frac{1}{(n+1)\{Y_{i+1,n} - Y_{i,n}\}} \quad (5.3)$$

constitutes a (large) sample estimate of $\tau(i/(n+1))$. By the strong law of large numbers, it follows from (5.2) that

$$\tau_X\left(\frac{i}{n+1}\right) \rightsquigarrow \frac{1}{(n+1)\{E(Y_{i+1,n}) - E(Y_{i,n})\}}. \quad (5.4)$$

Note, now, the following expression for the difference of means of order statistics in (5.4):

$$E(Y_{i+1,n}) - E(Y_{i,n}) = \binom{n}{i} \int_{-\infty}^{\infty} F_X^i(x) \{1 - F_X(x)\}^{n-i} dx, \quad (5.5)$$

for $i = 1, \dots, n-1$. ((5.5) is proved in the appendix.) Similarly, by substitution of $u = F_X(x)$, the following variant on (4.3) is obtained:

COROLLARY

$$\frac{(n+1)!}{i!(n-i)!} \int_0^1 \frac{1}{\tau_X(u)} u^i (1-u)^{n-i} du \rightsquigarrow \frac{1}{\tau_X\left(\frac{i}{n+1}\right)}. \quad (5.6)$$

Note that (4.3) and (5.6) are beta integrals for $\tau_X = 1$. Also τ_X and $1/\tau$ asymptotically 'factor out' of these beta-type integrals as $n \rightarrow \infty$.

6. Some Inferential Applications

6.1 Location and Scale Parameters

Suppose that $-\infty < \mu < \infty$ and $\sigma > 0$ are location and scale parameters of X , respectively, i.e. the distribution function, F_Z , of $Z = (X - \mu)/\sigma$ is independent of both μ and σ with

$$F_X(x) = F_Z\left(\frac{x - \mu}{\sigma}\right) \quad (6.1)$$

and thus

$$F_Z^{-1}(F_X(x)) = \frac{x - \mu}{\sigma}. \quad (6.2)$$

Differentiation of (6.2) with respect to x and (2.6) yield that

$$\tau_X(u, \sigma) = \frac{1}{\sigma} \tau_Z(u), \quad (6.3)$$

where τ_Z is independent of μ and σ . From (6.3) it follows that τ_X is independent of μ , which is also clear from location invariance of density ordinates.

Next, consider theorem (2.5) with regard to the likelihood function,

$$L_X(\mu, \sigma) = \prod_{i=1}^n f_X(X_i) = \prod_{i=1}^n \xi_{i,n}. \quad (6.4)$$

From (2.5) and (6.3), it is clear that a first-order approximation to the likelihood function is given by $\sigma^{-n} \Pi \tau_Z(i/(n+1))$. In addition, if $\ell_X(\mu, \sigma) = -\log L_X(\mu, \sigma)/n$, the so-called 'support' function, then by the strong law of large numbers,

$$\ell_X(\mu, \sigma) \rightarrow H_X(\sigma), \quad (6.5)$$

almost surely, as $n \rightarrow \infty$, where $H_X(\sigma) = -E\{\log f_X(X)\}$, the 'entropy' of X . It is easily seen that H_X is independent of μ and, in particular $H_X(\sigma) =$

$\log \sigma + H_Z$, where $H_Z = -E\{\log f_Z(Z)\}$, the entropy of Z , a parameter independent of μ and σ . Consequently, we have the

COROLLARY:

$$T_X(n) = -\frac{1}{n} \sum_{i=1}^n \log \tau_X\left(\frac{i}{n+1}\right) \rightarrow H_X = \log \sigma + H_Z \quad (6.6)$$

as $n \rightarrow \infty$. (Only the latter part (6.6) is exact.)

It is interesting to note from (6.6) that as $n \rightarrow \infty$, the likelihood function 'loses' its dependence on μ before its dependence on σ and that it breaks down into terms involving scale and entropy in its approach to 0.

6.2 An Estimation Procedure

Theorem (4.1) suggests a likelihood based estimation procedure for estimating an unknown parameter, θ of F_X . It consists of equating the likelihood function, $L_X(\theta) = \prod_{i=1}^n f_X(x_i, \theta)$, with $\prod_{i=1}^n \tau_X(i/(n+1))$, effectively providing a sample independent likelihood ordinate in terms of τ_X . This procedure may be given by

$$\ell_X(\hat{\theta}) = T_X(n, \hat{\theta}), \quad (6.7)$$

where $\hat{\theta}$ denotes the sample estimate and T_X is defined in (6.6). It also follows from (6.6) that as $n \rightarrow \infty$ the estimator, (6.7) approaches $H_X(\hat{\theta})$ almost surely. This is the same as the frequency moment procedure suggested in Manas (1997, (Equation (4) page 95)), which gives maximum likelihood estimates in exponential families. In the examples, it is seen that, for finite n , (6.7) generalizes the maximum likelihood approach by altering the sample size, n , to correct for bias.

7. Examples

1. EXPONENTIAL(θ) DISTRIBUTION:

Suppose that $F_X(x, \theta) = 1 - e^{-\theta x}$, and hence $f_X(x, \theta) = \theta e^{-\theta x}$, for $x > 0$, where $E(X) = 1/\theta > 0$ is a scale parameter. It follows that $\tau_X(u) = \theta(1-u)$, for $0 < u < 1$, so that for this X , τ_X is linearly decreasing on $(0, 1)$. In addition (4.2) is exact and (6.6) becomes

$$T_X(n) = -\frac{1}{n} \sum_{i=1}^n \log \frac{i}{n+1} \rightarrow 1, \quad (7.1)$$

as $n \rightarrow \infty$.

The estimation procedure, (6.7), becomes

$$\hat{\theta} = \frac{T_X(n)}{\bar{X}}, \quad (7.2)$$

since $\ell_X(\theta) = -\log \theta + \theta \bar{X}$, where $\bar{X} = \sum X_i/n$. Note that since $T_X(n)$ approaches 1 from below, (7.2) is partially correcting the upward bias in the maximum likelihood estimator, $1/\bar{X}$, whose exact mean is $n\theta/(n-1)$.

2. PARETO(α) DISTRIBUTION:

Let $F_X(x, \alpha) = 1 - (1+x)^{1-\alpha}$ so that $f_X(x, \alpha) = (\alpha-1)(1+x)^{-\alpha}$, where $x > 0$ and $\alpha > 1$ is a 'shape' parameter. It follows that $\tau_X(u, \alpha) = (\alpha-1)(1-u)^{\alpha/(\alpha-1)}$, for $0 < u < 1$. From τ_X , it can be seen that a Pareto(α) population approaches exponential($\alpha-1$) as $\alpha \rightarrow \infty$. With this τ_X and $\rho = \alpha/(\alpha-1) > 1$, (4.3) becomes

$$\frac{n!}{(n-i)!} \frac{\Gamma(n-i+\rho+1)}{\Gamma(n+\rho+1)} \sim \left(1 - \frac{i}{n+1}\right)^\rho, \quad (7.3)$$

and (5.6) becomes the variant

$$\frac{\Gamma(n+2)}{\Gamma(n-\rho+2)} \frac{\Gamma(n-i-\rho+1)}{\Gamma(n-i+1)} \sim \left(1 - \frac{i}{n+1}\right)^{-\rho}. \quad (7.4)$$

To illustrate the rate of convergence to τ_X , Table 1 provides values of (5.3) and the reciprocal of (5.4) for $\alpha = 3$, i.e. $\rho = 1.5$ and various i, n .

Table 1. Approximations to τ_X in a Pareto(3) distribution

n	i	$\{(n-i+1)/(n+i)\}^{1.5}$	Approx. in (5.3)	Approx. in (5.4)
4	1	0.7155	0.7272	0.4374
20	5	0.6650	0.6686	0.5992
40	10	0.6574	0.6594	0.6238
200	50	0.6511	0.6515	0.6443

Application of (6.6) yields (7.1) as in the case of the exponential distribution.

The estimation procedure, (6.7) becomes

$$\hat{\alpha} = \frac{T_X(n)}{\log \bar{G}} + 1, \quad (7.5)$$

with T_X defined in (7.1), and $\bar{G} = \left\{ \prod_{i=1}^n (X_i + 1) \right\}^{1/n}$, the geometric mean of $X_i + 1$ over $i = 1, \dots, n$, a sufficient statistic for α . Since this is an exponential family, (7.5) gives the maximum likelihood estimator, $1/\log \bar{G} + 1$ as $n \rightarrow \infty$.

3. CAUCHY(σ) DISTRIBUTION:

Let $f_X(x, \sigma) = \left\{ \sigma \pi \left(1 + (x/\sigma)^2 \right) \right\}^{-1}$ and $F_X(x, \sigma) = \frac{1}{2} + \pi^{-1} \arctan(x/\sigma)$ for $-\infty < x < \infty$ and $\sigma > 0$ a scale parameter. From well-known properties of trigonometric functions, it follows that $\tau_X(u) = (\sigma\pi)^{-1} \sin^2(\pi u)$ for $0 < u < 1$. It was shown in Manas (1997) that the entropy of this distribution is $H_X(\sigma) = \log(4\pi\sigma)$ and hence from (6.6),

$$-\frac{1}{n} \sum_{i=1}^n \log \left\{ \sin \left(\frac{i\pi}{n+1} \right) \right\} \rightarrow \log 2, \quad (7.6)$$

as $n \rightarrow \infty$.

4. LOGISTIC DISTRIBUTION:

Consider the standard logistic distribution, with $F_Z(z) = (1 + e^{-z})^{-1}$ for $-\infty < z < \infty$. It follows that $\tau_Z(u) = u(1 - u)$, a quadratic function on $(0, 1)$. In this case, the limiting results in (4.3) and (5.6) are obvious and (6.6) effectively yields (7.1) since $H_Z = 1 + 1 = 2$.

8. Concluding remarks

From the examples, we see a striking simplicity in the form of the functions τ_X , an alternate form of the density function. Clearly a closed form for τ_X may exist at the percentiles even though no closed form exists for F_X . This is the case with the normal, gamma, Weibull, etc. distributions.

There are also practical applications to density estimation and simulation. In particular, (5.3) is reminiscent of the so-called 'empiric p.d.f.', given by

$$\hat{f}_X(x) = \frac{2}{n \{Y_{i+1, n} - Y_{i-1, n}\}},$$

for $i = 2, \dots, n - 1$ an estimator of f_X , based on a continuous empirical distribution function, obtained by joining with straight lines successive midpoints of the empirical distribution function (see e.g. Dudewicz and Mizra, 1988).

With regard to simulation, it is clear that for fixed n , percentiles satisfying (4.1) give large sample order statistics from F_X without integration. Large samples of X -values could thus easily be generated using (4.1) from which random samples could be drawn, providing an alternative to 'Monte Carlo' simulation. Comparative studies on the advantages of this method of simulation are currently under investigation.

9. Appendix

PROOF OF THEOREM (4.1):

($\Rightarrow$) If x satisfies $F_X(x) = i/(n+1)$, then $x = F_X^{-1}(i/(n+1))$. Thus

$$f_X(x) = f_X(F_X^{-1}(i/(n+1))) = \tau_X(i/(n+1)).$$

($\Leftarrow$) We have seen that for each $i = 1, \dots, n$ there exists an x_i for which

$f_X(x_i) = \tau_X(i/(n+1)) = f_X(F_X^{-1}(i/(n+1)))$, where $x_1 < x_2 < \dots < x_n$. Thus $x_i = F_X^{-1}(i/(n+1))$ and hence $F_X(x_i) = i/(n+1)$.

PROOF OF (5.5):

The distribution function of $Y_{i,n}$ is

$$F_{Y_{i,n}}(x) = \sum_{k=i}^n \binom{n}{k} F_X^k(x) \{1 - F_X(x)\}^{n-k} \quad (\text{A.1})$$

and thus $F_{Y_{i,n}}$ satisfies the recurrence

$$F_{Y_{i,n}}(x) = \binom{n}{i} g_{i,n}(x) + F_{Y_{i+1,n}}(x), \quad (\text{A.2})$$

where $g_{i,n}(x) = F_X^i(x) \{1 - F_X(x)\}^{n-i}$.

Upon differentiation, (A.2) becomes

$$f_{Y_{i,n}}(x) = \binom{n}{i} g'_{i,n}(x) + f_{Y_{i+1,n}}(x), \quad (\text{A.3})$$

with $f_{Y_{i,n}}$ given in (4.5). Multiplying (A.3) by x and integrating over $-\infty < x < \infty$ gives

$$E(Y_{i+1,n}) - E(Y_{i,n}) = - \binom{n}{i} \int_{-\infty}^{\infty} x g'_{i,n}(x) dx. \quad (\text{A.4})$$

Applying integration by parts to the integral in (A.4) gives

$$\int_{-\infty}^{\infty} x g'_{i,n}(x) dx = - \int_{-\infty}^{\infty} g_{i,n}(x) dx \quad (\text{A.5})$$

and the result follows.

References

- BARNETT, V. (1975). Probability plotting methods and order statistics. *Appl. Statist.*, **24**, 95–108.
- DAVID, H.A. (1981). *Order Statistics*, Second ed. Wiley, New York.
- HARTER, H.L. (1984). Another look at plotting positions. *Commun. Statist.*, **13**, 1613–1633.
- HARTER, H.L. (1988). History and role of order statistics. *Commun. Statist. - Theory Meth.*, **17**(7), 2091–2107.
- HARTER, H.L. & WEIGAND, R.P. (1985). A Monte Carlo study of plotting positions. *Commun. Statist.*, **B14**, 317–343.
- MANAS, G.J. (1997). A new estimation procedure based on frequency moments. *South African Statist. J.*, **31**, 91–106.
- DUDEWICZ, E.J. & MISHRA, S.N. (1988). *Modern Mathematical Statistics*. Wiley, New York.

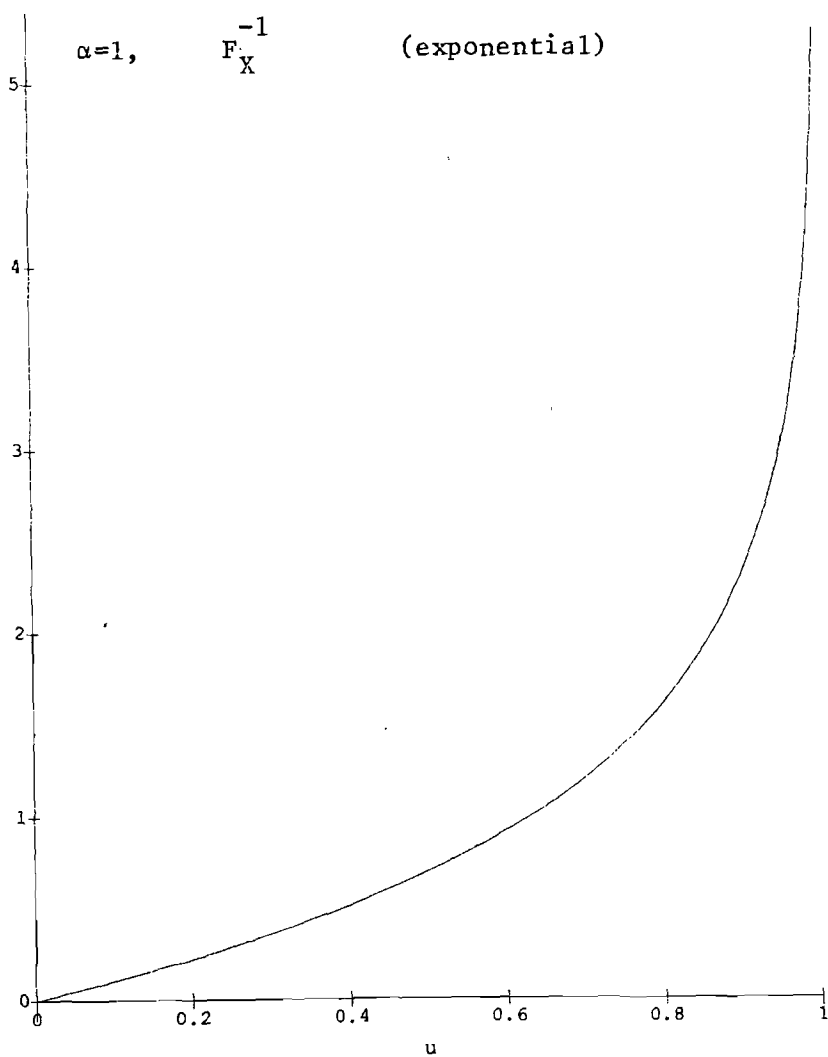
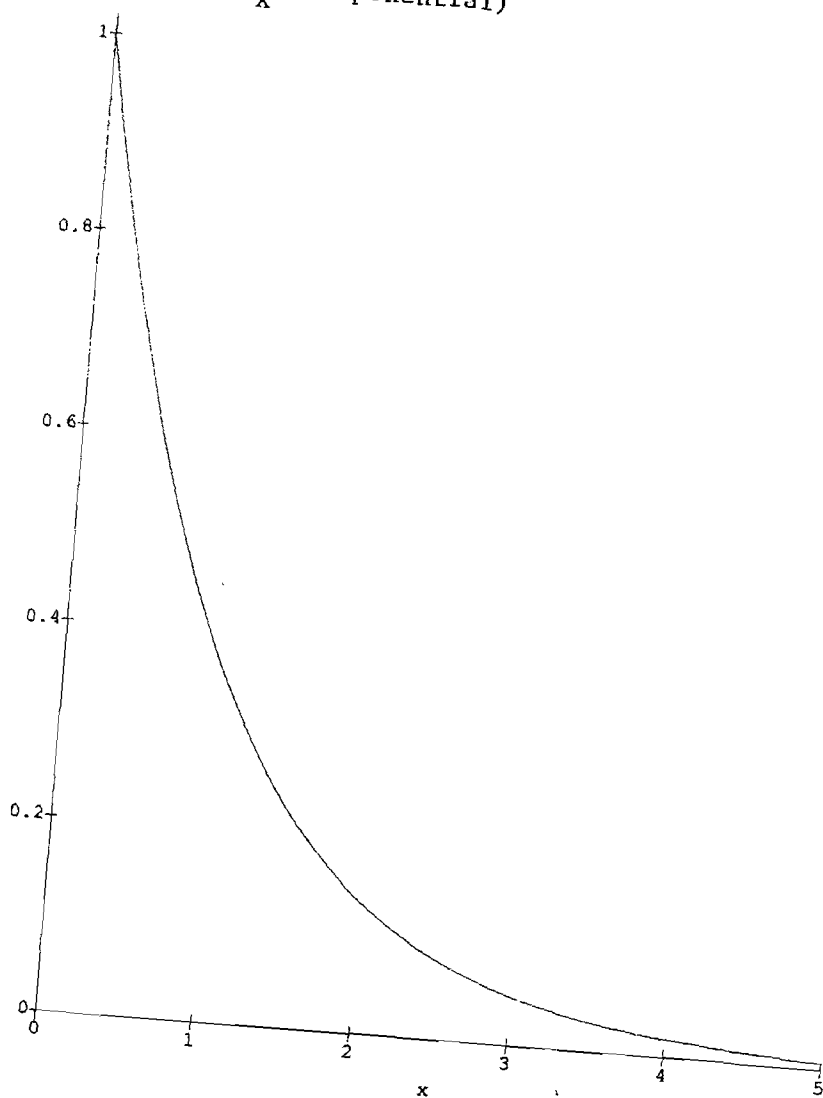
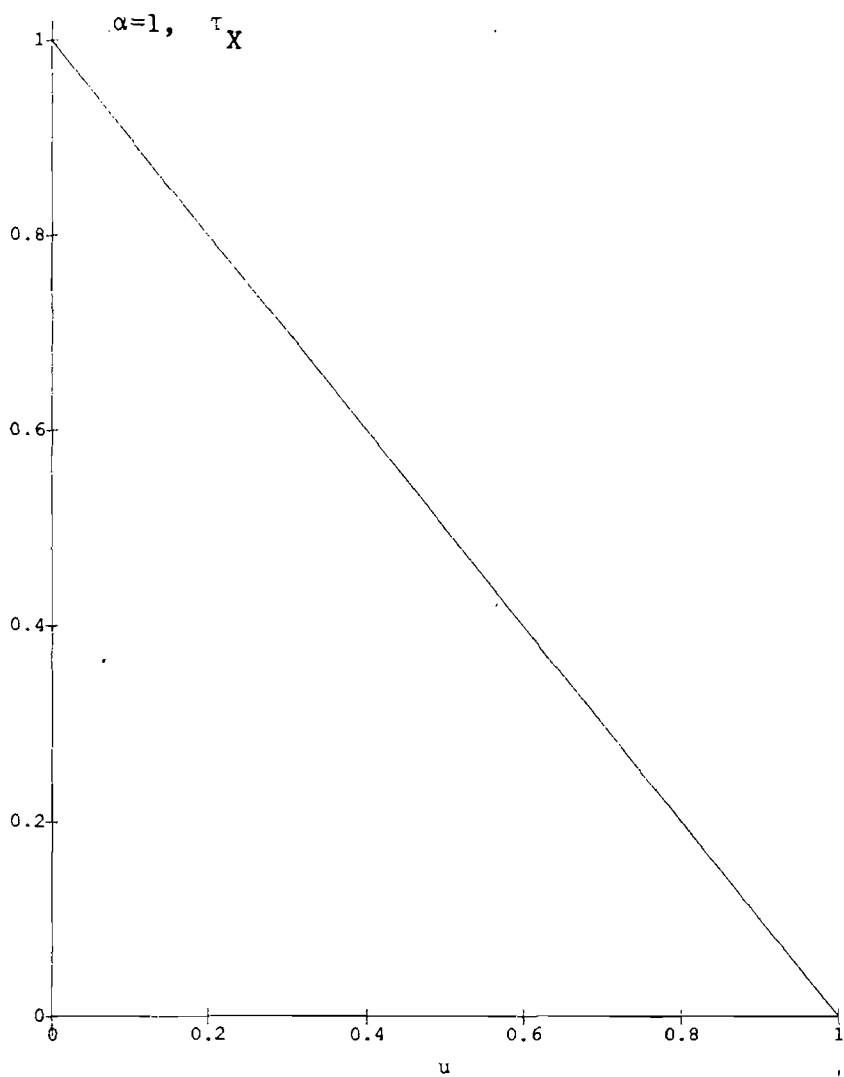


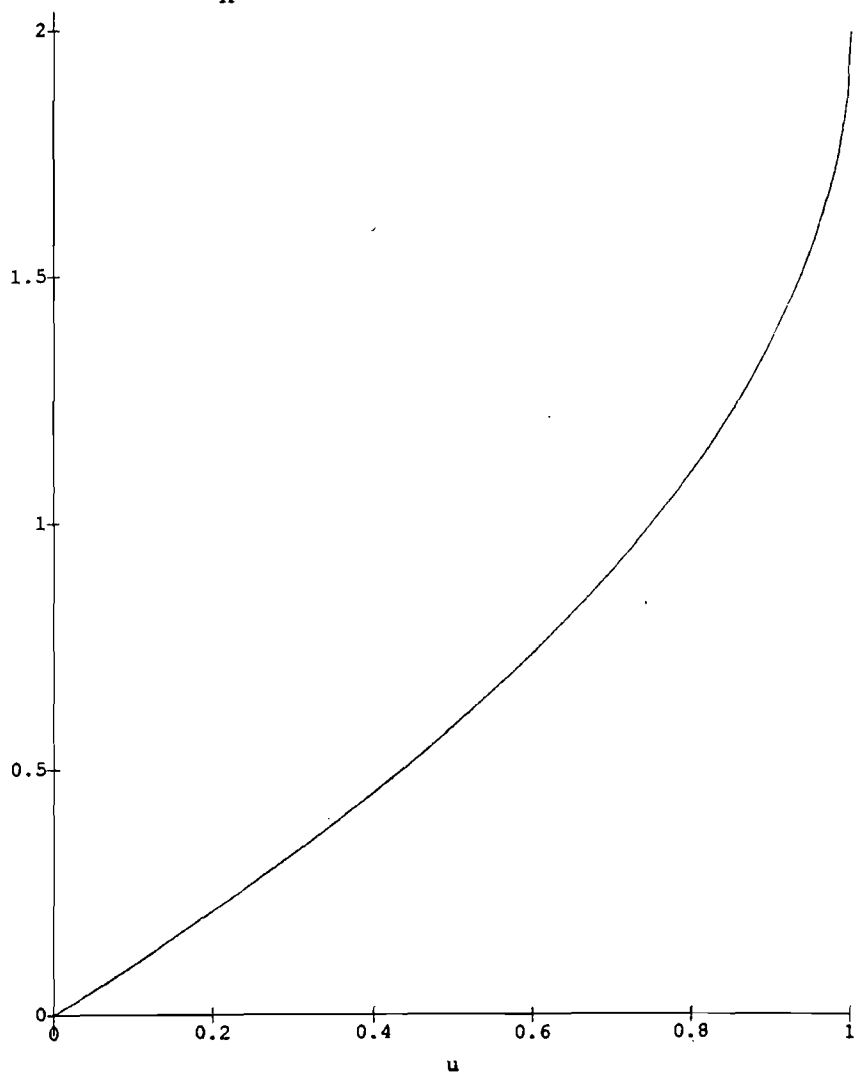
Figure 1. Graphs of F_X^{-1} , f_X and τ_X for $\alpha = 0.5, 1$ and 2 , respectively.

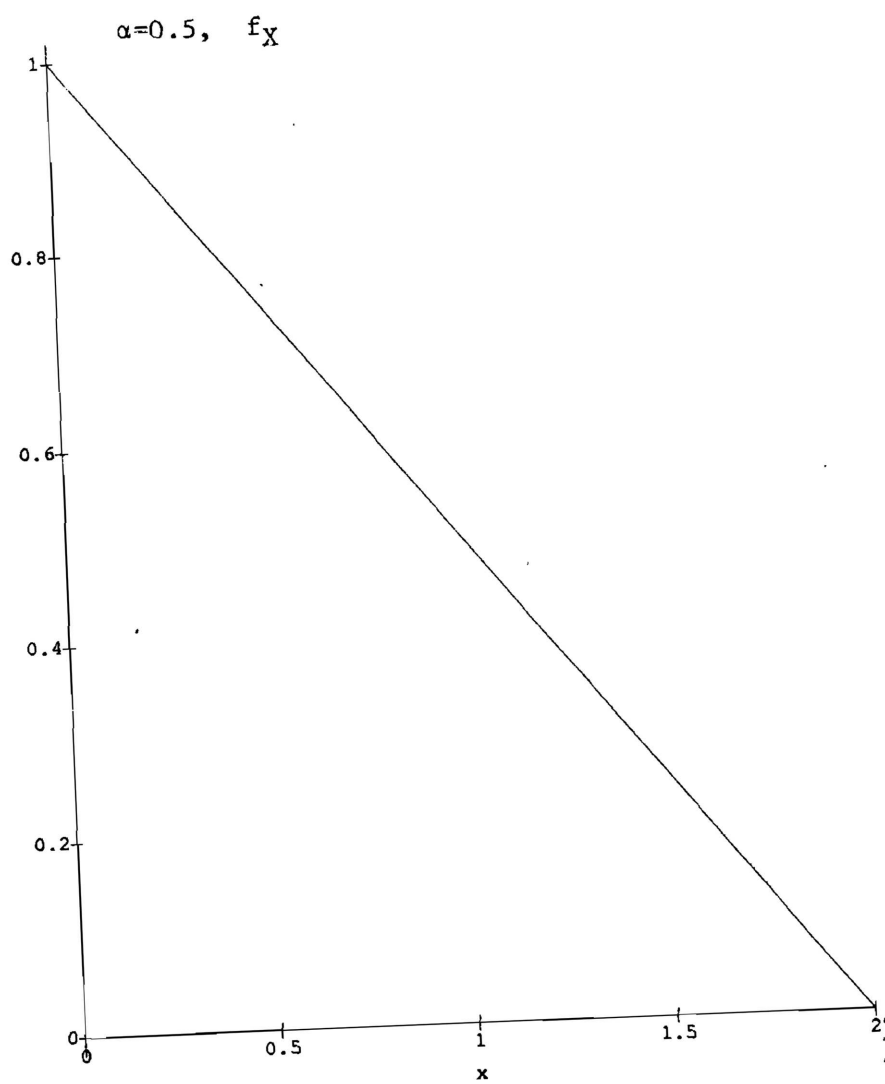
$\alpha=1, f_X$ (exponential)

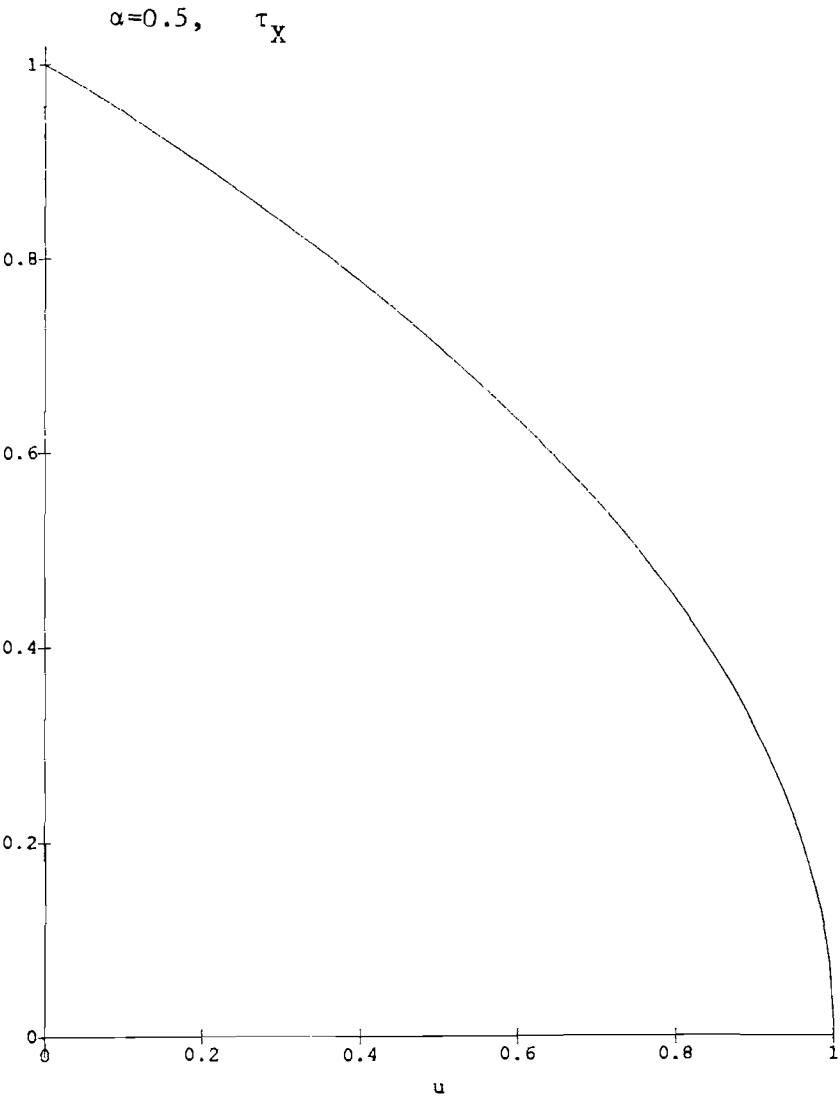




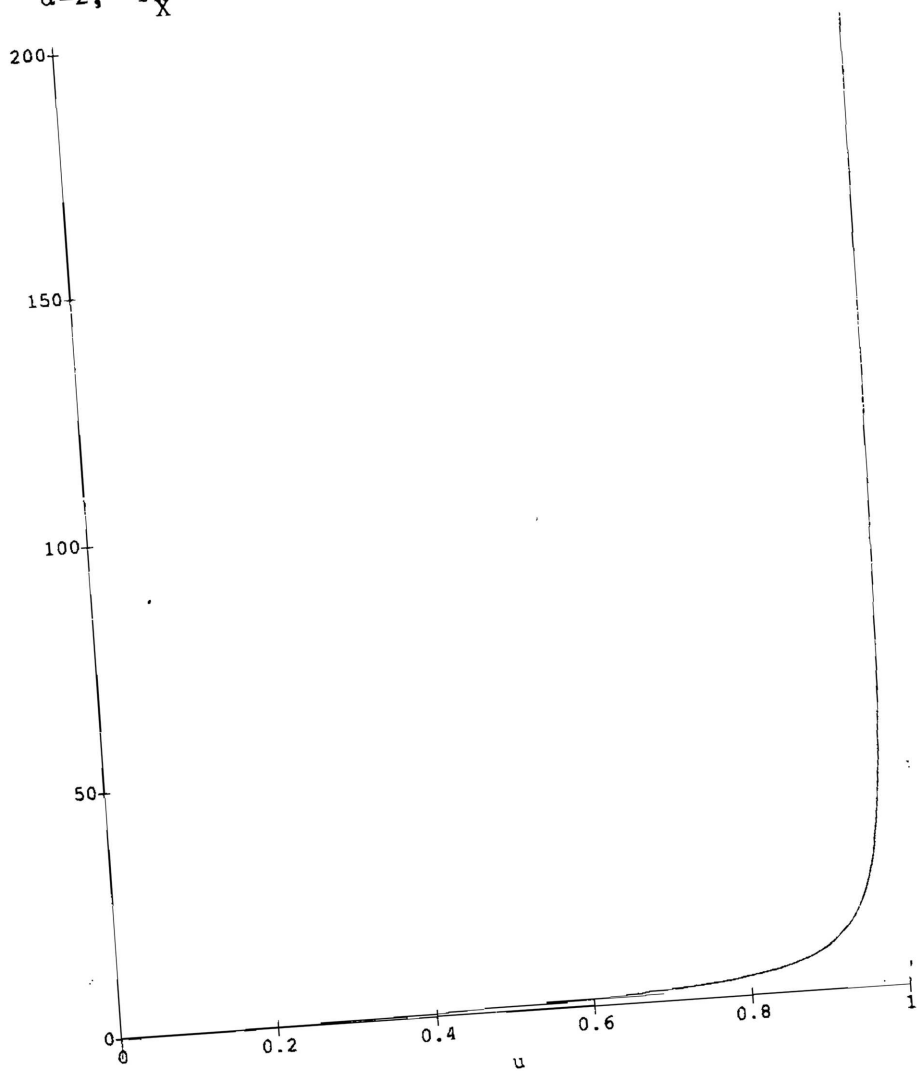
$$\alpha=0.5, F_X^{-1}$$







$$\alpha=2, \quad F_X^{-1}$$



DENSITY ORDINATES OF POPULATION PERCENTILES

145

$$\alpha=2, \quad f_X$$

